

Instrumental Variables

IV FRONTIERS



Roadmap

Introductions

IV Interpretation

- LATE and Generalizations

- Characterizing Compliers

- Diff-in-Diff and IV

Formula Instruments

- Shift-Share IV

- Recentered IV

Chinese Economic Applications: Session II

Concluding Thoughts

Session 2: Overview

Welcome back! In Session 1, we covered:

- IV mechanics: just-identified IV, 2SLS, weak instruments
- Applications: ADH (2013) China Syndrome and BDQ (2020) China roads

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- Applications: ADH (2013) China Syndrome and BDQ (2020) China roads

Session 2 agenda:

1. **IV Interpretation:** What does IV *really* identify?
 - LATE, compliers, treatment effect heterogeneity
 - Connections to diff-in-diff
2. **Formula Instruments:** Shift-share and recentered IV
 - Formal conditions, implementation, China HSR illustration
3. **Chinese Economics Applications: Session II**
 - Meng, Qian, Yared (2015, ReStud): Causes of China's Great Famine

Review: Key Takeaways from Session 1

Three IV estimands/estimators to know:

- Just-identified IV = ρ/π : reduced form over first stage
- 2SLS: optimal combination of multiple instruments; avoids many-IV bias if few instruments used
- Wald estimand: for binary Z , $= E[Y|Z = 1] - E[Y|Z = 0]$ over $E[D|Z = 1] - E[D|Z = 0]$

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Two papers illustrated the “double-level” logic for instrument validity:

- **ADH (2013)**: Chinese import shocks in other countries $\rightarrow$ shift-share IV
- **BDQ (2020)**: 1962 historical plan $\rightarrow$ straight-line IV for Chinese roads

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A key question we did *not* answer: under what conditions does β^{IV}

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LATE and Generalizations

Characterizing Compilers

Diff-in-Diff and IV

Formula Instruments

Shift-Share IV

Recentered IV

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What Does IV Identify, Really?

IV was invented in the context of structural economic models, typically with a single parameter β linearly relating Y_i to X_i

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The Imbens-Angrist LATE result revolutionized our understanding of IV estimands, and clarified some subtle points around IV identification

- β^{IV} often identifies a convex average of heterogeneous effects under first-stage *monotonicity*: Z_i only affects X_i in one direction
- IV “exogeneity” can arise from two conceptually different assumptions of instrument *independence* and *exclusion*

Basic (Binary Treatment, Binary Instrument) Setup

Let $Y_i(0)$ and $Y_i(1)$ denote individual i 's potential outcomes given a binary treatment $D_i \in \{0, 1\}$

- Observed outcomes: $Y_i = (Y_i(1) - Y_i(0)) D_i + Y_i(0)$

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Under what assumptions can we causally interpret `ivreg2 Y (D=Z)?`

Imbens and Angrist (1994) Assumptions

1. *As-good-as-random assignment*: $Z_i \perp (Y_i(0), Y_i(1), D_i(0), D_i(1))$

→ Consider the Angrist draft lottery, or Angrist-Krueger's QoB IV

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 - Equivalently, given Assumption 1, $E[D_i(1) - D_i(0)] \neq 0$
4. *Monotonicity*: $D_i(1) \geq D_i(0)$ for all i (i.e., almost-surely)
 - The instrument can only shift the treatment in one direction

Local Average Treatment Effect (LATE) Identification

Imbens and Angrist showed that under these assumptions:

$$\beta^{IV} = E[Y_i(1) - Y_i(0) \mid D_i(1) > D_i(0)]$$

The IV estimand β^{IV} identifies a LATE: the average treatment effect $Y_i(1) - Y_i(0)$ among *compliers*: those with $1 = D_i(1) > D_i(0) = 0$

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- Monotonicity rules out the presence of *defiers*, with $D_i(1) < D_i(0)$

IV Estimates of the Effects of Military Service on the Earnings of White Men born in 1950

Earnings year	Earnings		Veteran Status		Wald Estimate of Veteran Effect
	Mean	Eligibility Effect	Mean	Eligibility Effect	
	(1)	(2)	(3)	(4)	(5)
1981	16,461	-435.8 (210.5)	.267	.159 (.040)	-2,741 (1,324)
1971	3,338	-325.9 (46.6)			-2050 (293)
1969	2,299	-2.0 (34.5)			

Note: Adapted from Table 5 in Angrist and Krueger (1999) and author tabulations. Standard errors are shown in parentheses. Earnings data are from Social Security administrative records. Figures are in nominal dollars. Veteran status data are from the Survey of Program Participation. There are about 13,500 individuals in the sample.

What Does This Mean *Practically*?

Two conceptually distinct considerations: *internal* vs. *external* validity

- Context of an IV, and who the compliers likely are, may matter
- Usual “overidentification” test logic fails: two valid IVs may have different estimands

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- Sensible in many settings (e.g. draft lottery / cancer screening RCTs)
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Note: technically, we don't need monotonicity if effects are homogenous

Generalizations and Limitations

The core logic of IA'94 extends to multivalued treatments/instruments

- IV identifies an avg. of incremental treatment effects, putting more weight on margins where the instrument shifts the treatment more
- Regressions of $\mathbf{1}[X_i \geq x]$ on Z_i identify the weights

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Since 2SLS is a weighted average of “one-at-a-time” IVs, overidentified single-treatment specifications can have a LATE interpretation

- Need all individual IVs to have a LATE interpretation and the 2SLS weights to be convex (the latter can be checked empirically)

Generalizations and Limitations (Cont.)

We can also have a LATE interpretation in IV specifications w/controls

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Key condition: the controls are flexible enough to make $E[Z_i | W_i]$ linear

- E.g. W_i is a set of mutually exclusive strata dummies
- IV then identifies a weighted-average of conditional-on- W_i IVs

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If $E[Z_i | W_i]$ is nonlinear, IV may be biased (even w/constant effects)

- Key practical takeaway: it's good to check sensitivity to how controls are parameterized (add interactions, higher-order polynomials, etc.)

Who Are the Compliers?

Characterizing the i that make up the IV estimand ($w/D_i(1) > D_i(0)$) is key for understanding internal vs. external validity

- Unfortunately we can't identify compliers directly: we only observe $D_i(1)$ (when $Z_i = 1$) or $D_i(0)$ (when $Z_i = 0$), not both together!

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As it turns out, we can still characterize compliers by their outcomes ($Y_i(0)$ and $Y_i(1)$) and by other observables X_i

- Comparing $E[X_i \mid D_i(1) > D_i(0)]$ to $E[X_i]$ can shed light on how $E[Y_i(1) - Y_i(0) \mid D_i(1) > D_i(0)]$ compares to $E[Y_i(1) - Y_i(0)]$

Outcomes

Computing $E[Y_i(1) \mid D_i(1) > D_i(0)]$ is surprisingly easy in the IA setup

- Define $W_i = Y_i D_i$, and note that this new outcome has potentials with respect to D_i of $W_i(1) = Y_i(1)$ and $W_i(0) = 0$

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- Thus IV with W_i as the outcome identifies

$$E[W_i(1) - W_i(0) \mid D_i(1) > D_i(0)] = E[Y_i(1) \mid D_i(1) > D_i(0)]$$

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$$E[W_i(1) - W_i(0) \mid D_i(1) > D_i(0)] = E[Y_i(1) \mid D_i(1) > D_i(0)]$$

Similar logic shows that IV with $Y_i(1 - D_i)$ as the outcome and $1 - D_i$ as the treatment identifies $E[Y_i(0) \mid D_i(1) > D_i(0)]$

- So easy to do! And extends to covariates / multiple IVs...

Illustration: Angrist et al. (2013) Charter School IV

Subject	Urban				Nonurban			
	Treatment effect (1)	$E_u[Y_0 D=0]$ (2)	λ_0^u (3)	λ_1^u (4)	Treatment effect (5)	$E_n[Y_0 D=0]$ (6)	λ_0^n (7)	λ_1^n (8)
<i>Panel A. Middle school</i>								
Math	0.483*** (0.074)	-0.399*** (0.011)	0.077 (0.049)	0.560*** (0.054)	-0.177** (0.074)	0.236*** (0.007)	0.010 (0.061)	-0.143*** (0.042)
N	4,858				2,239			
ELA	0.188*** (0.064)	-0.422*** (0.012)	0.118** (0.054)	0.306*** (0.049)	-0.148*** (0.048)	0.260*** (0.007)	0.102** (0.050)	-0.086*** (0.030)
N	4,551				2,323			

Decomposing

$$\begin{aligned}
 LATE = & \underbrace{E[Y_i(1) \mid D_i(1) > D_i(0)] - E[Y_i(0) \mid D_i = 0]}_{\lambda_1} \\
 & - \underbrace{(E[Y_i(0) \mid D_i(1) > D_i(0)] - E[Y_i(0) \mid D_i = 0])}_{\lambda_0}
 \end{aligned}$$

shows that charter compliers have typical counterfactual achievement

Covariates

For covariates X_i (not affected by D_i) we can follow a similar trick:

- Either IV'ing $X_i D_i$ on D_i or IV'ing $X_i(1 - D_i)$ on $1 - D_i$ identifies complier characteristics $E[X_i \mid D_i(1) > D_i(0)]$
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Can be useful to compare with always- and never-taker means:

- AT: $E[X_i \mid D_i(1) = D_i(0) = 1] = E[X_i \mid D_i = 1, Z_i = 0]$
- NT: $E[X_i \mid D_i(1) = D_i(0) = 0] = E[X_i \mid D_i = 0, Z_i = 1]$

Illustration: Angrist et al. (2023) Charter School IV – Characteristics

	Compliers			Always-takers	Never-takers
	Untreated (1)	Treated (2)	Pooled (3)		
Female	0.506 (0.023)	0.510 (0.021)	0.508 (0.016)	0.539 (0.024)	0.463 (0.017)
Black	0.401 (0.022)	0.380 (0.021)	0.390 (0.016)	0.623 (0.023)	0.490 (0.017)
Hispanic	0.250 (0.02)	0.300 (0.018)	0.275 (0.013)	0.183 (0.019)	0.228 (0.014)
Asian	0.022 (0.007)	0.024 (0.005)	0.023 (0.004)	0.004 (0.003)	0.024 (0.005)
White	0.229 (0.018)	0.216 (0.016)	0.223 (0.012)	0.154 (0.016)	0.215 (0.014)
Special education	0.190 (0.018)	0.181 (0.016)	0.186 (0.012)	0.158 (0.018)	0.177 (0.013)
English language learner	0.143 (0.015)	0.148 (0.013)	0.145 (0.010)	0.054 (0.011)	0.088 (0.010)
Subsidized lunch	0.689 (0.021)	0.705 (0.019)	0.697 (0.014)	0.698 (0.022)	0.666 (0.016)
Baseline math score	-0.274 (0.047)	-0.312 (0.041)	-0.293 (0.032)	-0.394 (0.045)	-0.301 (0.036)
Baseline English score	-0.352 (0.050)	-0.349 (0.043)	-0.350 (0.033)	-0.362 (0.046)	-0.299 (0.038)

Fancier Things

General result: $E[g(X_i, Y_i(d)) \mid D_i(1) > D_i(0)]$ for any $g(\cdot)$ and any $d \in \{0, 1\}$ is identified by β in the IV regression:

$$g(X_i, Y_i) \times \mathbf{1}[D_i = d] = \alpha + \beta \mathbf{1}[D_i = d] + \varepsilon_i$$

$$\mathbf{1}[D_i = d] = \mu + \pi Z_i + \nu_i$$

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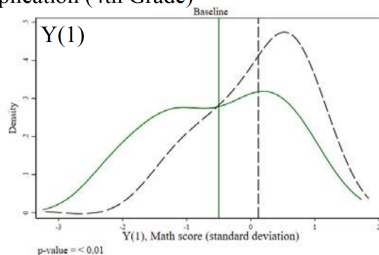
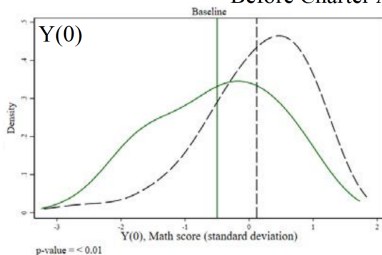
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Lots of fun stuff we can do here. E.g.: distributions

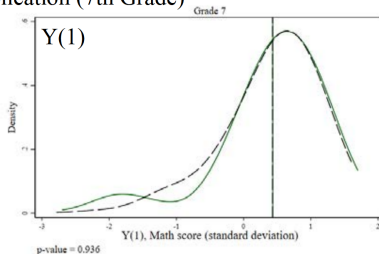
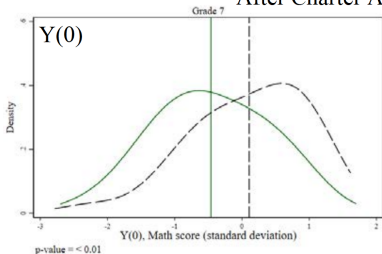
- $g(X_i, Y_i) = \mathbf{1}[Y_i \leq y]$ estimates the CDF of complier potential outcomes, $F(y) = Pr(Y_i(d) < y \mid D_i(1) > D_i(0))$
- $g(X_i, Y_i) = \frac{1}{h} K(\frac{Y_i - y}{h})$ estimates the corresponding PDF, where $K(\cdot)$ is a kernel function and h is a bandwidth

Illustration: Angrist et al. (2023) Charter School IV – Distributions

Before Charter Application (4th Grade)



After Charter Application (7th Grade)



Outside the Basic IA Setup

The same logic applies to IV regressions with flexible controls

- E.g. *ivreging* $Y_i D_i$ on D_i instrumenting by Z_i and controlling for cell FE ID's a weighted avg of within-cell $E[Y_i(1) \mid D_i(1) > D_i(0)]$

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Logic also goes through with continuous instruments

- E.g. can always mechanically decompose an IV on a binary D_i into implied " $Y(1)$ " and " $Y(0)$ " terms
- Unfortunately, things get trickier with non-binary treatments

What about Diff-in-Diff?

Most LATE-style discussions of IV identification are “design-based”:
i.e., the instrument is assumed to be drawn randomly, as if in an RCT

- But not all instruments are easily thought of this way (e.g. distance)

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Luckily, whenever you have a reduced form & first stage that are causal
and exclusion+monotonicity hold, the IV ratio has a LATE interpretation

- E.g. diff-in-diff IV, for distance measure $Z_i \in \{0, 1\}$ and $t \in \{1, 2\}$:

$$Y_{it} = \alpha_i + \tau_t + \beta D_{it} + \epsilon_{it}$$

$$D_{it} = \mu_i + \lambda_t + \pi Z_i \times \mathbf{1}[t = 2] + \epsilon_{it}$$

Parallel trends in Z_i make the reduced form & first stage causal

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- Recent TWFE literature becomes relevant with fancier specifications

Example: Duflo (2001) School-Building Diff-in-Diff

TABLE 3—MEANS OF EDUCATION AND LOG(WAGE) BY COHORT AND LEVEL OF PROGRAM CELLS

	Years of education			Log(wages)		
	Level of program in region of birth			Level of program in region of birth		
	High (1)	Low (2)	Difference (3)	High (4)	Low (5)	Difference (6)
<i>Panel A: Experiment of Interest</i>						
Aged 2 to 6 in 1974	8.49 (0.043)	9.76 (0.037)	-1.27 (0.057)	6.61 (0.0078)	6.73 (0.0064)	-0.12 (0.010)
Aged 12 to 17 in 1974	8.02 (0.053)	9.40 (0.042)	-1.39 (0.067)	6.87 (0.0085)	7.02 (0.0069)	-0.15 (0.011)
Difference	0.47 (0.070)	0.36 (0.038)	0.12 (0.089)	-0.26 (0.011)	-0.29 (0.0096)	0.026 (0.015)
<i>Panel B: Control Experiment</i>						
Aged 12 to 17 in 1974	8.02 (0.053)	9.40 (0.042)	-1.39 (0.067)	6.87 (0.0085)	7.02 (0.0069)	-0.15 (0.011)
Aged 18 to 24 in 1974	7.70 (0.059)	9.12 (0.044)	-1.42 (0.072)	6.92 (0.0097)	7.08 (0.0076)	-0.16 (0.012)
Difference	0.32 (0.080)	0.28 (0.061)	0.034 (0.098)	0.056 (0.013)	0.063 (0.010)	0.0070 (0.016)

Here Z_i = growing up in a region with intensive school building

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Concluding Thoughts

Exogenous Shocks with Non-Random Exposure

Increasingly, researchers construct instruments from multiple sources of variation — only some of which is viewed as exogenous. E.g.:

- Shift-share instruments $Z_i = \sum_k s_{ik}g_k$, which average together a set of shocks g_k with predetermined exposure shares s_{ik}
- IVs capturing spillovers across social networks/geographies
- “Simulated” instruments, capturing eligibility for a public program

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Generically, $Z_i = f_i(g, s)$ where $g = (g_1, \dots, g_k)$ is a set of shocks, s is some measure of shock exposure, and $f_i(\cdot)$ is a known formula

- How can we leverage exogeneity in g , allowing s to be non-random?

Example: Autor, Dorn, and Hanson (2013/2014)

ADH study the effects of rising Chinese import competition on US commuting zones in the 1990's and 2000's

- Treatment X_i : growth of Chinese imports in CZ i
- Main outcome Y_i : change in manufacturing employment in CZ i

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Idea: Z_i predicts X_i with proxies for underlying productivity shocks

- Fairly intuitive ... but what do we need for it to work?

SSIV with Exogenous Shocks

Borusyak, Hull, and Jaravel (2022) show shift-share IV can work when

- The shocks g_k are exogenous, at least conditional on some q_k
- There is enough independent variation in the shocks (i.e. large K)

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- The shocks g_k are exogenous, at least conditional on some q_k
- There is enough independent variation in the shocks (i.e. large K)

Intuitively, $Z_i = \sum_k s_{ik} g_k$ works as a “translation device,” bringing the k -level variation to bear on causal effects at the i level

- E.g. in ADH, an industry-level natural experiment can be used to estimate effects at the commuting zone level
- The shares s_{ik} used for this translation do not need to be exogenous!

Two Practical Considerations

Need to control for $E[Z_i \mid s, q] = E[\sum_k s_{ik} g_k \mid s, q] = \sum_k s_{ik} E[g_k \mid q]$
in order to isolate the as-good-as-random variation in shocks

- When $E[g_k \mid q] = q'_k \gamma$, this means controlling for $\sum_k s_{ik} q_k$
- E.g. in ADH, to isolate within-period shock variation, need to control for $\sum_k s_{ik} \times (\text{period FE})$

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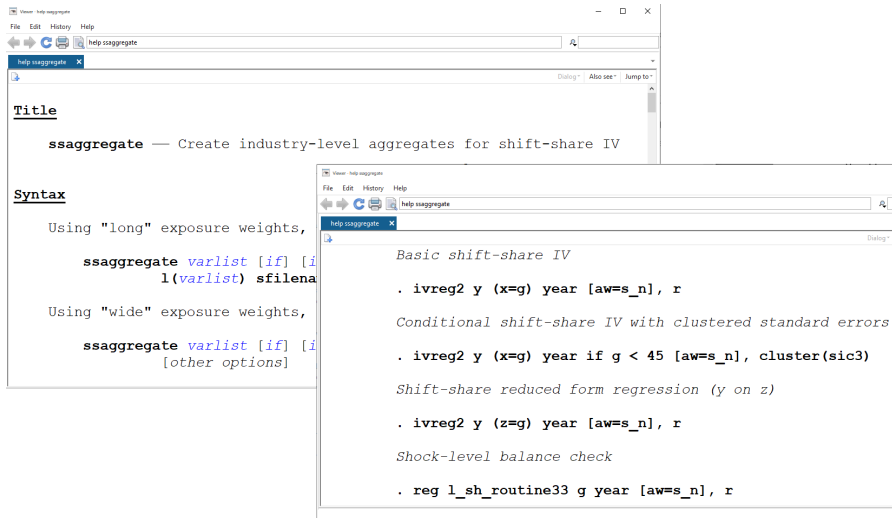
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Need to adjust SEs for a non-standard clustering: $Z_i = \sum_k s_{ik} g_k$ and $Z_j = \sum_k s_{jk} g_k$ are correlated via their common exposure to shocks

- Easy fix: use the *ssaggregate* Stata/R packages to translate SSIV estimation to the level of identifying variation (shocks)

Estimating Exogenous-Shock SSIV Regressions in Stata



Viewer: help ssaggregate

File Edit History Help

help ssaggregate

help ssaggregate

Title

ssaggregate — Create industry-level aggregates for shift-share IV

Syntax

Using "long" exposure weights,

```
ssaggregate varlist [if] [i] l(varlist) sfilena
```

Using "wide" exposure weights,

```
ssaggregate varlist [if] [i] [other options]
```

Viewer: help ssaggregate

File Edit History Help

help ssaggregate

help ssaggregate

Basic shift-share IV

```
. ivreg2 y (x=g) year [aw=s_n], r
```

Conditional shift-share IV with clustered standard errors

```
. ivreg2 y (x=g) year if g < 45 [aw=s_n], cluster(sic3)
```

Shift-share reduced form regression (y on z)

```
. ivreg2 y (z=g) year [aw=s_n], r
```

Shock-level balance check

```
. reg l_sh_routine33 g year [aw=s_n], r
```

To install: `ssc install ssaggregate`

Autor et al. (2013) Revisited

TABLE 4
Shift-share IV estimates of the effect of Chinese imports on manufacturing employment

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Coefficient	−0.596 (0.114)	−0.489 (0.100)	−0.267 (0.099)	−0.314 (0.107)	−0.310 (0.134)	−0.290 (0.129)	−0.432 (0.205)
Regional controls							
Autor <i>et al.</i> (2013) controls	✓	✓	✓		✓	✓	✓
Start-of-period mfg. share	✓						
Lagged mfg. share		✓	✓	✓	✓	✓	✓
Period-specific lagged mfg. share			✓	✓	✓	✓	✓
Lagged 10-sector shares					✓		✓
Local Acemoglu <i>et al.</i> (2016) controls						✓	
Lagged industry shares							✓
SSIV first stage <i>F</i> -stat.	185.6	166.7	123.6	272.4	64.6	63.3	27.6
No. of region-periods	1,444	1,444	1,444	1,444	1,444	1,444	1,444
No. of industry-periods	796	794	794	794	794	794	794

- Columns 3-7 include the key $\sum_k s_{ik} \times$ (period FE) control
- Standard errors / first stage *F*-stat.'s computed via *ssaggregate*

Beyond Shift-Share

Suppose we wanted to estimate the economic impact of new transportation upgrades (e.g. new high speed rail construction)

- Economic theory tells us to expect spillovers: easier travel increases productivity across all cities i in a country, to different extents
- E.g. *market access*: $X_i = \sum_j \tau(g, loc_i, loc_j)^{-1} pop_j$ where g captures the railroad network, loc_i gives lat/lon of cities, and pop_i is city size

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Like a more complex shift-share: $Z_i = f_i(g, s)$ for $s = (loc_i, pop_i)_{i=1}^N$

- Suppose we had a natural experiment in high-speed rail construction; can we “translate” these g shocks via the market access mapping?

Recentered IV

Borusyak and Hull (2023) show how to leverage exogenous g in $f_i(g, s)$:

1. Specify *counterfactual* exogenous shocks $g^{(1)}, \dots, g^{(C)}$ which may well have occurred (e.g. shuffle the timing of new rail construction)
2. Recompute the instrument: $Z_i^{(c)} = f_i(g^{(c)}, s)$, for $c = 1, \dots, C$
3. Take the average (“expected instrument”): $\mu_i = \frac{1}{C} \sum_i Z_i^{(c)}$

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Either controlling for μ_i or using the recentered $\tilde{Z}_i = Z_i - \mu_i$ avoids any bias from endogeneity/non-randomness in Z_i

- Analogous to controlling for $\sum_k s_{ik} q_k$ in shift-share IV!
- BH also discuss the analog of non-standard SSIV clustering: the solution is a bit more non-standard (randomization inference)

Illustration: High-Speed Rail in China

149 lines were built or planned (as of April 2019)

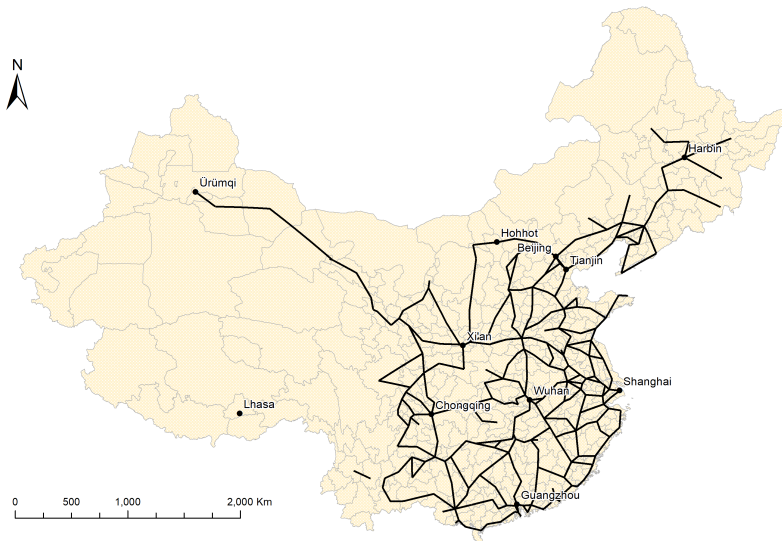


Illustration: High-Speed Rail in China

The 83 lines actually built by 2016. Suppose exact timing is random

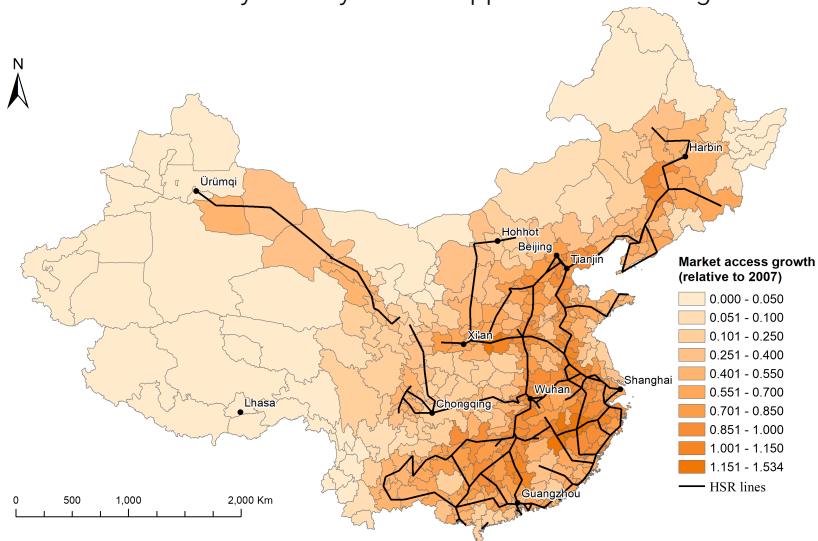


Illustration: High-Speed Rail in China

A counterfactual draw of 83 lines by 2016

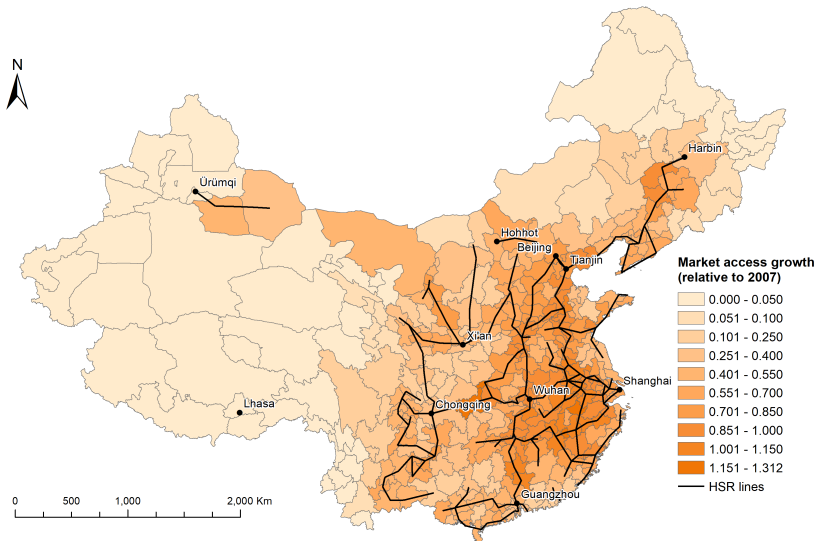


Illustration: High-Speed Rail in China

Expected MA growth, μ_i

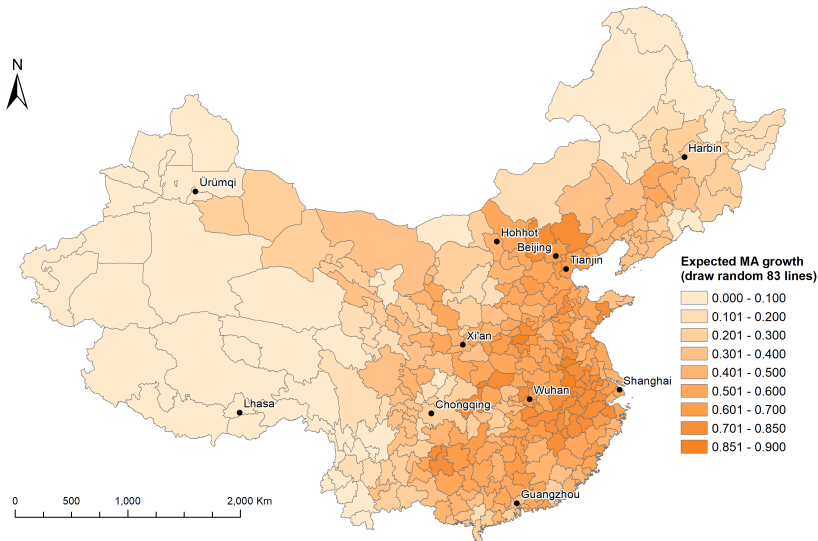
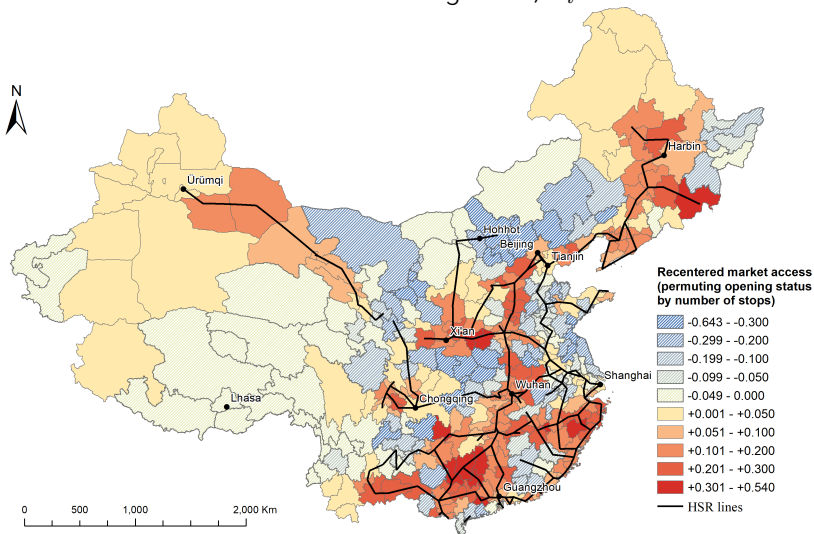


Illustration: High-Speed Rail in China

Recentered MA growth, $\tilde{Z}_i$



Adjusted Estimates of Market Access Effects

	Unadjusted OLS (1)	Recentered IV (2)	Controlled OLS (3)
<i>Panel A. No Controls</i>			
Market Access Growth	0.232 (0.075)	0.081 (0.098) [-0.315, 0.328]	0.069 (0.094) [-0.209, 0.331]
Expected Market Access Growth			0.318 (0.095)
<i>Panel B. With Geography Controls</i>			
Market Access Growth	0.132 (0.064)	0.055 (0.089) [-0.144, 0.278]	0.045 (0.092) [-0.154, 0.281]
Expected Market Access Growth			0.213 (0.073)
Recentered	No	Yes	Yes
Prefectures	274	274	274

Regressions of log employment growth on log market access growth in 2007–2016.
 Spatial-clustered standard errors in parentheses; permutation-based 95% CI in brackets

Recentering for Power

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Ex: Medicaid eligibility as a treatment combining statewide policy shocks and individual exposure (income, family structure, etc)

- In settings where policy shocks are plausibly exogenous, standard approach is to use them directly as instruments (“simulated IV”)
- BH approach: use eligibility itself, but recenter: e.g. adjust for i ’s avg. eligibility across permutations of policies (swap MA & RI, say)

Illustration: ACA Medicaid Expansion

	Has Medicaid		Has Private Insurance		Has Employer-Sponsored Insurance	
	Simulated IV (1)	Recentered IV (2)	Simulated IV (3)	Recentered IV (4)	Simulated IV (5)	Recentered IV (6)
<i>Panel A. Eligibility Effects</i>						
Eligibility	0.132 (0.028) [0.080,0.216]	0.072 (0.010) [0.051,0.093]	-0.048 (0.023) [-0.110,0.009]	-0.023 (0.007) [-0.040,-0.007]	0.009 (0.014) [-0.034,0.052]	-0.009 (0.005) [-0.021,0.004]
<i>Panel B. Enrollment Effects</i>						
Has Medicaid			-0.361 (0.165) [-0.813,0.082]	-0.321 (0.092) [-0.566,-0.108]	0.068 (0.111) [-0.232,0.421]	-0.125 (0.061) [-0.263,0.070]
P-value: SIV=RIV			0.719		0.104	
Exposed Sample	N	Y	N	Y	N	Y
States	43	43	43	43	43	43
Individuals	2,397,313	421,042	2,397,313	421,042	2,397,313	421,042

1% ACS sample of non-disabled adults in 2013–14, diff-in-diff IV regressions using one of the two instruments. Controls include state and year fixed effects and an indicator for Republican governor interacted with year. State-clustered standard errors in parentheses; wild score bootstrap 95% CI in brackets

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IV Methods in Chinese Economic History

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- The Great Leap Forward (1958–62): radical collectivization and grain procurement policies created quasi-exogenous variation in famine severity
- One-child policy: regional variation in enforcement creates IVs for fertility
- SOE reforms: variation in privatization timing and intensity
- Financial repression: capital controls and credit allocation

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In this section, we examine a landmark paper using IV to study one of the most consequential episodes in modern Chinese history:

- **Meng Qian, Yared (2015, ReStud):** The institutional causes of

Paper 3: “The Institutional Causes of China’s Great Famine”

Citation: Meng, X., Qian, N., and Yared, P. (2015). “The Institutional Causes of China’s Great Famine, 1959–1961.” *Review of Economic Studies*, 82(4): 1568–1611.

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Why it matters: Understanding the cause has implications for policy

MQY 2015: Research Question and Setting

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Key data challenge: Historical Chinese provincial statistics were suppressed or manipulated during the famine period — requires careful

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- Use **rainfall and temperature** as instruments for actual grain production
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(2) Grain production shock as IV for procurement:

- The grain production shock (instrumented by weather) shifts procurement quotas
- Allows estimation of the effect of *procurement* on mortality, holding

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- Illustrates IV in a **historical/archival** setting with imperfect data
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Summary: Chinese Economics Papers Using IV

Paper	Question	IV	Finding
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Some general advice on how to approach IV in the “real world”:

1. Figure out what the reduced form and first stage regressions are, & what variation in the instrument(s) they leverage via the controls
2. Assess the plausibility of as-good-as-random assignment (or parallel trends) of this variation, both theoretically and empirically
3. Interrogate exclusion, which lets you interpret RF/FS, and monotonicity/complier stats if heterogeneous effects seem

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Further Resources

Textbooks and reviews:

- Angrist and Pischke (2009): *Mostly Harmless Econometrics*
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Methodological frontiers:

- Borusyak, Hull, Jaravel (2022): Shift-share IV formal conditions
- Borusyak and Hull (2023): Recentered IV
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China-specific:

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If you can answer all three convincingly, you likely have a credible IV design — the foundation of publishable empirical work

Keep Calm and ivreg2 On!

Thank you for your attention throughout both sessions!

Jingxuan Du

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Good luck on your future adventures with IV!